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The Issues of Selection Warehouse Process Strategies

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Abstract

Paper presents problems of designing and organization logistic processes in warehouse facilities. Highlighted the need of improving logistics facilities actions as a key aspect of increasing productivity, flexibility and reliability of supply chains. For this purpose, warehouse process and the most common sub-processes and activities included in it were characterized. Additionally, it was noted that, warehouse processes may be implemented in many different ways and under various strategies. Selection of particular warehouse strategy usually depends on basic warehouse tasks, structure and size of customer orders, handled logistic units, costs of materials handling in particular sub-processes (both financial and time costs), availability of storage space and labour resources, etc. Therefore, probability of selecting the particular strategy is dependent on technical and organizational factors. Because of that, the paper presents some approach to selecting warehousing strategy. This approach takes into account mentioned technical and organizational factors. It also uses decision-making tree which shows probabilities of selection subsequent warehouse operations (in particular moment and warehouse situation) as well as probability of such situations as e.g. shipment suspending or cancelation in case of choosing a given strategy. Examples of such decision-making tree with probabilities of warehouse strategy selection is shown in paper.

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1. Introduction

Increasing globalization, development of world economies and growing consumerism of societies leads to increase demand for transport, movement and logistic handling of material goods. It causes creation of new as well as expansion and increasing complexity of existing logistics systems and supply chains. Consequently, they are expanded with new logistics facilities performing a wide range of logistics tasks. These tasks are aimed at efficient and effective transformation of materials in such way to satisfy customer needs (with rationalization of incurred costs at the same time). This in turn determines the necessity of solving a number of decision-making problems regarding the designing of efficient and effectively manage supply chains [7], [8].

The role of logistics facilities is a crucial for all supply chains, because they have significant contribution to handle materials which are moved from the places of production to consumers. Correct operation of warehouse facilities enables the realization of their logistic tasks at appropriate and acceptable by customers quality level. Consequently, it determines validity, cost-effectiveness and need of their functioning in supply chains. Therefore, a lot of attention in the literature is given to issues related to the design of storage facilities (see e.g. [1], [2], [4], [5], [6]) as well as modelling and organization of their warehouse processes (see e.g. [3], [9], [10], [11], [12]).

2. Warehouse process

Flow of materials through all kinds of logistics facilities is a strictly defined sequence of transformations performed on these materials. These transformations may involve transformations of time, place or form of handled materials. In the first case, it concerns buffering and storage of materials in logistic facilities. Place transformation is understood as a movement and transport of materials within warehouse. However, during the form transformation, materials are processed due to their physical form, i.e., co-packed, consolidated, unconsolidated, assembled, packaged, etc.

Warehouse process is a set of actions which are associated with receiving, storage, picking and shipping of material goods, in a suitably adapted places for this purpose, and under certain organizational and technological conditions. Therefore it can be concluded that warehouse process includes such sub-processes as receiving, storage, picking and shipping. Nevertheless, it is a very general approach, and warehouse process may take many different forms, and include multiple sub-components. Selection and appropriate connection of this process elements (sub-processes) is determined by functions and tasks of logistics facility. In fact, production warehouses, distribution warehouses or crossdocking warehouses usually perform variety kinds of transformations.

Each of these warehouse process components is characterized by an appropriate sequence of actions that have to be performed to complete a given goals and objectives (see Table 1). It is closely related to materials and information transformation by labour resources (labour resources include warehouse employees, transport means, warehouse equipment, tools for information flow management, etc.).

Table 1. The most common sub-processes and activities included in warehouse process [8].

Sub-process	Activity	Transformation
1	2	3
Receiving	Unloading	place
	Cargo identification and control	time
	Buffering	time
Put-away	Transport to storage area	place
	Placing unit loads in storage location	place
Storage	–	time
Replenishment	Transport to order picking area	place
	Transformation of unit loads to form offered in order picking	form
	Replenishment to pick locations	place

End of the Table 1

1	2	3
	Placing remaining unit loads in storage location	place
	Placing in selected location empty bins (pallets)	place
Order picking	Replenishment to pick locations	place
	Preparing items for picking	form
	Picking items	place
	Sorting, packing, preparing picked unit loads for transport	form
	Transport of prepared unit loads to selected place in order picking area	place
	Transport of prepared unit loads to buffer	place
Co-packing	Transport of unit loads to co-packing stations	place
	Preparing items for co-packing	form
	Creation new SKU (e.g. promotional SKU sets, combined SKU)	form
	Packaging, labelling, tagging, foiling, etc.	form
	Transport of prepared unit loads to selected place of buffer	place
Consolidation, deconsolidation, sortation	–	form (quantity)
Shipping	Buffering	time
	Cargo identification and control	time
	Loading	place
Material return policy, utilization	Cargo identification and control	time
	Buffering	time
	Loading	place
Crossdocking	Transport from input buffer to output buffer	place

3. Warehouse process strategy selection

Warehouse process can be implemented in many different ways (under different strategies). It means that materials can be processed by various components of warehouse process, in their various configurations as well as in various functional areas of logistics facility (see Fig. 1). Therefore, it can be assumed, that warehouse process strategy is a part of this process. It determines the sequence of warehouse operations. These operations are related to internal transport, cargo form transformation and cargo buffering between their input and output from the warehouse.

Selection of particular warehouse strategy usually depends on basic warehouse tasks, structure and size of customer orders, handled logistic units, costs of materials handling in particular sub-processes (both financial and time costs), availability of storage space and labour resources, etc.

As mentioned, the cost of handling cargo units by warehouse process strategy can be expressed both in money and time units. Implementation of particular warehouse operations absorbs as well money (correlated with engagement of labour resources, funds freezing, maintenance of infrastructure and suprastructure) as time required to perform them.

The total cost of material unit handling under particular strategy is the sum of the costs of individual activities (C_i) as well as the costs of movement between these operations and costs of delaying (C_{iw}). Costs of delaying are costs incurred when unit loads have to wait because some warehouse activities cannot be performed (eg. due to lack of storage space, lack of SKU to pick, lack of labour resource to perform the operations).

The total cost of material unit handling under particular strategy is the sum of the costs of performing each pair of consecutive operations and transport between them ($C_{i,j}$) and costs of unit loads delaying (C_{iw}). For example, total

cost of material handling under s-th strategy (C_s) which include operations {1, 2, 3, 4, 6, 10} (see Fig. 1) can be presented as follows:

$$C_s = C_{1,2} + C_{1w} + C_{2,3} + C_{2w} + C_{3,4} + C_{3w} + C_{4,6} + C_{4w} + C_{6,10} + C_{6w} \quad (1)$$

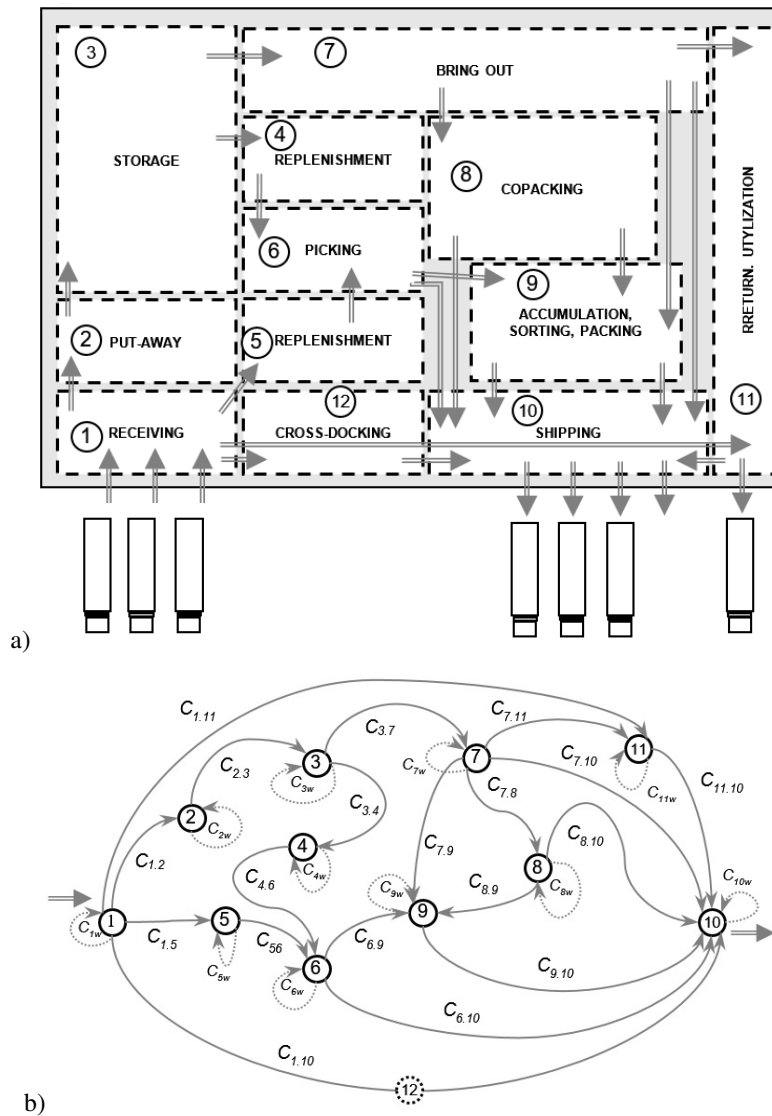


Fig. 1. Functional areas and sub-processes of warehousing (a); Graph of warehouse strategy costs (b).

Selection of the appropriate warehouse strategy will therefore be dictated by finding such a variant of material flow, in which the cost will be the lowest. At the same time it must be acceptable according to feasibility of the given logistics task (e.g. total time of transforming materials under a given strategy cannot exceed time of order processing, acceptable by customer).

However, selection of warehouse strategy depends not always on costs of its implementation. The main factor influencing this choice may be the availability of labour resources or, for example:

- in case of deliveries – free space in buffer and storage areas, the need to replenishment order picking area,
- in case of shipments – availability of ordered materials in different functional areas.

Therefore, probability of selecting the particular strategy is dependent on technical and organizational factors. Of course, selection of strategy that reflects a standard process is more likely than the other. If in some functional area is lack of the requested material or it will not be available in the expected picking time, the probability of choosing strategy that supports this area is 0. Similarly, probability will be equal 0, if in a particular functional area will not be enough space to store, buffering or handling received materials.

For each task of shipping or receiving it is possible to determine the decision-making tree, according to which could be selected, with a certain probability, strategy of warehouse process. Example of such a tree for a single shipping task is shown in Fig. 2.

In the example (shown in Fig. 2) shipping task implementation may cause with picking of material (green colour) shipment suspending (delay treated as an error handling – orange colour) or cancellation of task (causing losses, customer dissatisfaction – red colour). Selection of successive strategy components in decision tree is done with a certain probability, e.g.: path selection, taking into account the ability to pick from order picking area (relation 1–15) will take place with probability $P(X_{1,15})$:

$$P(X_{1,15}) = P(X_{3,4}) \cdot P(X_{4,15}) + P(X_{3,4}) \cdot P(X_{4,5}) \cdot P(X_{5,12}) + P(X_{3,4}) \cdot P(X_{4,5}) \cdot P(X_{5,6}) \cdot P(X_{6,7}). \quad (2)$$

The probability of completion shipping tasks on time (P_R) can be determined based on the equation (2):

$$P_R = P(X_{3,4}) \cdot P(X_{4,15}) + P(X_{3,4}) \cdot P(X_{4,5}) \cdot P(X_{5,12}) + P(X_{3,4}) \cdot P(X_{4,5}) \cdot P(X_{5,6}) \cdot P(X_{6,7}) + P(X_{3,9}) \cdot P(X_{9,16}) + P(X_{3,9}) \cdot P(X_{9,10}) \cdot P(X_{10,17}). \quad (3)$$

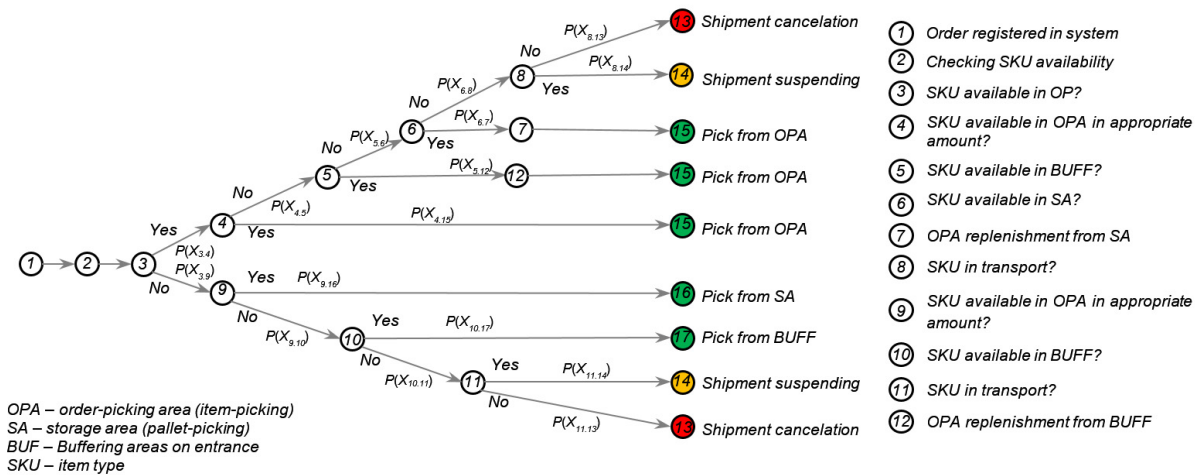


Fig. 2. Decision-making tree of strategy elements selection.

Determination of probability values to assign individual operations to warehouse strategy is based on a series of data analysis concerning logistics processes in a particular logistics facility. Estimating costs of whole warehouse process (taking into account specified warehouse organization, equipment, technology, labour resources availability, etc.) is possible by knowledge about material flow volumes, probability of selection particular strategy, and costs of handling one unit loads under particular strategy (Fig. 1b).

In addition, it is possible to predict losses that logistics facility may incur with a certain probability as a result of:

- shipment suspending P_S :

$$P_S = P(X_{3,4}) \cdot P(X_{4,5}) \cdot P(X_{5,6}) \cdot P(X_{6,8}) \cdot P(X_{8,14}) + P(X_{3,9}) \cdot P(X_{9,10}) \cdot P(X_{10,11}) \cdot P(X_{11,14}); \quad (4)$$

- shipment cancellation P_C :

$$P_C = P(X_{3,4}) \cdot P(X_{4,5}) \cdot P(X_{5,6}) \cdot P(X_{6,8}) \cdot P(X_{8,13}) + P(X_{3,9}) \cdot P(X_{9,10}) \cdot P(X_{10,11}) \cdot P(X_{11,13}). \quad (5)$$

Probability of selecting the particular strategy variant can be modified by e.g.: investments and modernization increasing technological potential of functional areas (performance, capacity) as well as by organizational activities related to allocation labour resources to tasks [10]. It should be noted that, e.g. increasing capacity of order picking area – OPA (see Fig. 2) will increase probability of finding needed SKU at this area. At the same time it can cause a prolongation of picking paths, and consequently leads to decreasing efficiency of shipping process. This indicates that all reorganizations, investment and technological changes in logistics facilities should be multifaceted analysed. Any such change may significantly affects the other components of warehouse strategy, the probability of strategy selection and thus the productivity and costs of warehouse process and whole facility.

4. Summary

The paper presents an approach to organizing and determining the sequence of activities performed on logistic units in warehouse facilities. Main attention was paid to what should be taken into consideration for selecting a specific strategy of materials handling. It includes not only costs that result from the way of materials handling, but also the time-consuming of warehouse operations, as well as other technological and organizational factors. These factors are e.g. availability of warehouse space or labour resources which will enable performance of individual storage operations in acceptable time and cost. Procedure of selection materials handling strategy was presented by using probabilities dependent on mentioned factors. It was also indicated that the value of the probability of selection of a given strategy can be modified by the relevant investments and reorganizations of warehouse areas.

Discussed issues are an introduction to constructing mathematical models for different variants (strategies) of materials transformations in logistics systems. Subsequently, it will allow to carrying out a wide range of analysis related to construction and organization of processes occurring such facilities. Based on them, logistics facilities can be evaluated in terms of performance and costs of warehouse processes, as well as their reliability (reliability of logistics facility is understood as a readiness of it to implementation of customer orders at specified time and on specified level of quality).

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